Measurement of Faraday Rotation Using Different Current Configurations

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INTRODUCTION

In 1865, James Clerk Maxwell presented his theory of electromagnetism, demonstrating that light is an electromagnetic wave consisting of electric and magnetic fields that propagate in the same direction, with both fields oscillating perpendicular to each other and to the direction of propagation [1]. Although the relationship between light and electromagnetism was not yet understood prior to Maxwell's work, experimental evidence had already been provided by Michael Faraday in 1845, who observed that magnetic fields can influence light as it passes through certain materials by rotating its polarization [2]. This phenomenon is known as Faraday rotation.

The polarization of light describes the evolution of its electric field vector over time and provides a reference for observing the rotation induced by a magnetic field. Since the magnetic field produces a negligible effect on the polarization of light propagating through air, a solenoid is used to generate a magnetic field within a material placed inside it. The field modifies the material's response to different components of the light's electric field, which results in a measurable rotation of the polarization, or Faraday rotation. The strength of Faraday rotation exhibited by a material in the presence of a magnetic field is quantified by its Verdet constant:

$$\nu(\lambda) = \frac{\pi}{\lambda}\frac{n^2(\lambda)-1}{n(\lambda)}\left(A + \frac{B}{\lambda^2-\lambda_0^2}\right)$$

where $n(\lambda)$ is the refractive index and A, B, and λ_0 are material-specific constants [3]. This relation was used to calculate the theoretical Verdet constant at $\lambda = 650$ nm.

In this experiment, polarized light from a laser diode is passed through an SF-57 glass rod subjected to a magnetic field to determine the Faraday rotation angle for different values of current through a solenoid. The Verdet constant of SF-57 is then determined and compared with the theoretical value of $\nu = 21.1\,\mathrm{rad}/(\mathrm{T}\cdot\mathrm{m})$ for SF-57 glass [3]. The experiment is repeated using different current configurations and measurement techniques, including the oscilloscope and lock-in amplifier, to compare the results and verify the accuracy of the measurements.

EXPERIMENTAL SETUP

This experiment focused on determining the Verdet constant of SF-57 glass from the rotation angle of polarization as a function of magnetic field strength generated by a solenoid. The true zero position of the adjustable polarizer was first determined using Malus' law by halving the sum of the rotation angles in either direction of the zero position to observe 80% of maximum intensity. The polarizer was then set to +45° relative to this position to maximize sensitivity to small rotations in polarization.

A current transformer wound with 10 turns of line was connected to the solenoid, which allowed for indirect measurements of the solenoid current and magnetic field strength by converting its current into a voltage signal that could be observed on the multimeter or oscilloscope. In determining the material's Faraday rotation angle and Verdet constant, three different experimental configurations were used:

- DC-Field (DC)
- AC-Field and Oscilloscope (OSC)
- AC-Field and Lock-in Amplifier (AMP)

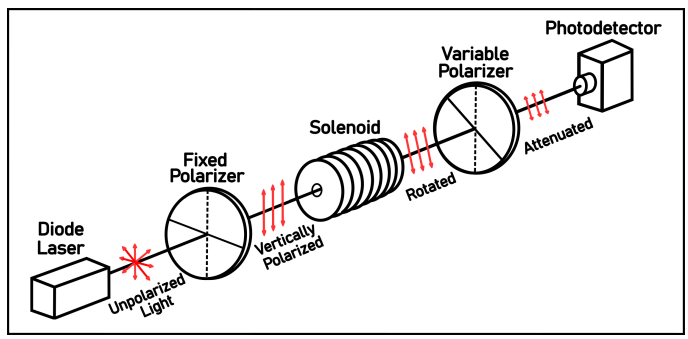

FIG. 1. Schematic of the optical setup used to observe the Faraday rotation of light emitted by a diode laser. Light is vertically polarized before passing through an SF-57 glass rod within a solenoid, where it undergoes Faraday rotation. The rotated polarization is attenuated by a variable polarizer set to +45° relative to the true zero position prior to detection by the photodetector.

For each experimental configuration, light from a 650 nm laser diode was passed through a fixed polarizer and then through an SF-57 glass rod of length 100 mm. The rod was placed within a solenoid calibrated to have a magnetic field of $B = (11.1\ \mathrm{mT/A})\ I_\mathrm{c}$, where I_c is the current through the solenoid. A variable polarizer was positioned between the solenoid and a photodetector to measure changes in light intensity resulting from Faraday rotation. The photodetector output was recorded as a voltage using a multimeter and oscilloscope. The strength of the magnetic field and corresponding Faraday rotation angle were then used to determine the Verdet constant of the material, with the accuracy of each method being assessed along with potential sources of noise associated with each configuration.

RESULTS

When polarized light is transmitted through a material in the presence of a magnetic field, the material responds differently to different components of the light's electric field. The light experiences a measurable angle of rotation θ depending on the magnetic field strength and the Verdet constant of the material, described by:

$$\theta = \nu B L \tag{1}$$

- ν is the Verdet constant of the material
- B is the magnetic field strength
- L is the length of the glass rod

The angle of rotation can also be approximated as:

$$\theta \approx \frac{\Delta I}{2 I_0} \tag{2}$$

- ΔI is the observed changed in intensity
- I_0 is the initial intensity

Using this angle and the predetermined magnetic field strength B, the Verdet constant of the material can then be determined for a specific solenoid current.

Section A verifies Malus' law using measurements obtained from adjusting the angle of rotation. Sections B, C, and D analyzes the different current configurations and measurement techniques used to obtain the angle of rotation from an observed change in intensity and assesses the accuracy of these methods in determining the Verdet constant for SF-57 glass.

A. Verifying Malus' Law

Malus' law is used to determine the final intensity of light transmitted through a polarizer as a function of the angle between the light's polarization and the transmission axis of the polarizer:

$$I(\phi, \theta) = I_0 \cos^2(\phi - \theta) \tag{3}$$

- I_M is the intensity of the incident light
- ϕ is the fixed angle of the polarizer
- θ is the angle of rotation

The true zero position was first determined using Malus' law in order to establish a reference angle and select an optimal value of ϕ for sensitivity to Faraday rotation. The polarizer was initially set to 0°, yielding a maximum voltage of 0.316 V. The true zero was then determined by rotating the polarizer in both directions until 80% of the maximum voltage (0.253 V) was measured on either side. The corresponding angles were -32° and +31°, which were averaged to obtain a true zero position of -0.5°.

From the true zero position, the polarizer was rotated through a total range of 120° in 5° increments, with the corresponding position and voltage recorded at each step. These measurements were plotted, and Eq. 1 was also applied to generate a theoretical curve describing the relationship between voltage and polarization angle. As shown in Fig. 2, this relationship exhibits a $\cos^2(\theta)$ dependence; symmetric about 0° and strictly nonnegative.

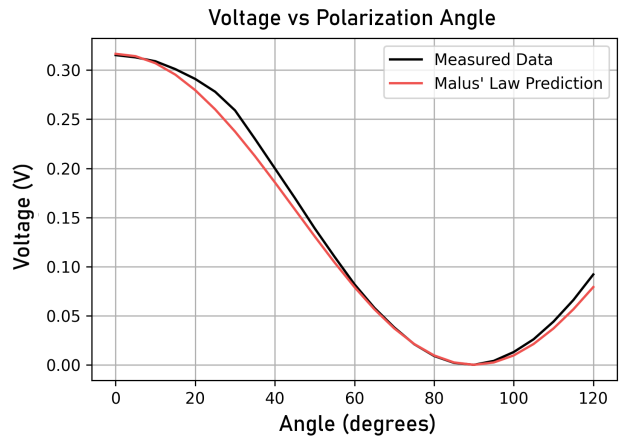

FIG. 2. Plot shows measured voltage values (black) along with the theoretical prediction from Malus' law (red). The measured data closely follows the expected $\cos^2\theta$ dependence.

After verifying that measurements aligned with the theoretical prediction from Malus' law, the polarizer was set to +45° relative to the true zero position of -0.5° for subsequent experiments, and no further adjustments of the polarizer were required.

B. Faraday Rotation due to a DC-Field

With 650 nm laser diode emitting into the SF-57 glass rod situated inside the coil, the initial voltage V_0 was first recorded from the photodetector to be 0.147 V in the absence of a magnetic field ($I_c = 0$). A DC magnetic field was then generated by varying the current through the solenoid via adjustments to the power supply voltage. The resulting Faraday rotation produced changes in the transmitted light intensity, which was measured as photodetector voltages V for five different current values.

Current (A)	Voltage (V)	θ_{rad}	θ_{deg}	Verdet Const.
1.0	0.142	0.017	1.0	15.3
1.5	0.138	0.031	1.8	18.4
2.0	0.136	0.037	2.1	16.9
2.5	0.132	0.051	2.9	18.4
3.0	0.131	0.054	3.1	16.3

TABLE I. Measured photodetector voltages and corresponding angle of polarization for different current values using a DC magnetic field. Verdet constants have units of rad/(T·M).

Substituting $\Delta I = \Delta V = V - V_0$ and $I_0 = V_0$ into Eq. 2 yielded the angle of rotation for each value of current, as shown in Table I. Using Eq. 1, the Verdet constants were found to range from 15.3 to 18.4, with a mean value of 17.1, differing by around 19% below the expected value of 21.1 rad/(T·M) for SF-57 glass. Fig. 4 illustrates the relationship θ vs B obtained from all three methods and is found in a later section.

C. AC-Field and Oscilloscope (OSC)

Previously, applying direct current to the solenoid generated a constant magnetic field with a constant voltage that was easily observed using a multimeter. However, when the solenoid is instead driven by alternating currents, this results in a time-varying magnetic field that oscillates in intensity and direction. For this current configuration, the photodetector signal is best measured using an oscilloscope.

An AC magnetic field was generated by driving a solenoid with an alternating current from the lock-in amplifier. The current amplitude was varied using the tuning function, which adjusted the peak-to-peak voltage of the output signal. Unlike the DC-field method, a voltage difference $V - V_0$ cannot be measured due to a constantly changing signal. Instead, the voltage oscillates around the baseline value corresponding to the fixed angle of the polarizer. The rotation angle can then be determined from the amplitude of this oscillation, as it represents the magnitude of the change in intensity.

Current (A)	Amplitude (V)	θ_{rad}	θ_{deg}	Verdet Const.
1.0	0.0066	0.02	1.29	20.2
1.5	0.0100	0.03	1.95	20.4
2.0	0.0130	0.04	2.53	19.9
2.5	0.0160	0.05	3.12	19.6
3.0	0.0195	0.07	3.80	19.9

TABLE II. SF-57 Glass Rod: Measured amplitude of oscillation signal and corresponding angle of polarization for different current values using an AC magnetic field. Verdet constants have units of rad/(T·M).

The amplitude A of the oscillation was obtained by halving the peak-to-peak voltage before substitution into Eq. 2 in place of ΔI. Using $I_0 = 0.147\ V$, the rotation angle was computed for each of the five current values, as shown in Table 2. After substituting this angle into Eq. 1, the Verdet constants were found to have a mean value of 20.0, with an error of around 5% from the expected value of 21.1 rad/(T·M) for SF-57 glass. Fig. 4 shows the relationship θ vs B and is located in a later section.

D. AC-Field and Lock-in Amplifier (AMP)

An AC magnetic field was generated by driving a solenoid with an alternating current from the lock-in amplifier, with the same signal also being used as a reference to set the phase that maximizes a response. When the voltage from the photodetector was multiplied by the reference signal, any signal components with matching frequency to the reference signal acquired a DC value, and components at other frequencies are shifted away before being removed by a low-pass filter to isolate the desired signal. The signal was averaged over a time constant of 300 ms after isolation, smoothening the output and effectively reducing noise from the oscillatory components.

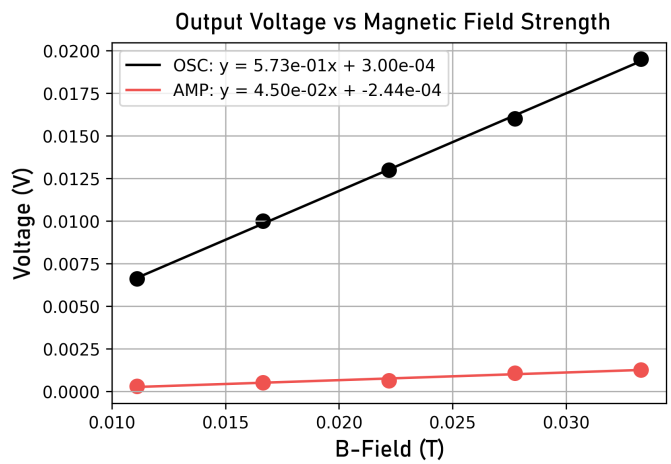

FIG. 3. Plot shows output voltage vs magnetic field strength for the two AC methods: Oscilloscope (OSC) and Lock-in Amplifier (AMP). Lines indicate the slopes obtained using regression analysis. Voltages obtained from the AMP method were divided by a gain of 10^4 for physical scaling.

The output voltage obtained from this method is proportional to the amplitude, but not necessarily equal to the true amplitude. Due to this disproportion, a scaling factor is required to relate the output voltage to the actual amplitude of the modulated signal. The ratio of the slopes V vs B obtained from the two AC methods (Fig 3) can be used to determine the scaling factor k:

$$k = \frac{dV_{\text{OSC}}/dB}{dV_{\text{AMP}}/dB} \tag{4}$$

Using the slope values of 0.573 and 0.045 for the OSC and AMP method respectively, the scaling factor was found to be 12.74. This factor converts the output from the lock-in amplifier into a peak-to-peak voltage equivalent to the measurement from the OSC method. Finally, the Verdet constant is determined using the relation:

$$\nu = \frac{1}{L} \cdot \frac{1}{2I_o} \cdot \frac{dV_{\text{AMP}}}{dB} \cdot k \tag{5}$$

After using the lock-in amplifier to reduce noise and isolate the signal, the Verdet constant of the SF-57 glass rod was found to be 19.5, with an error of 7.6% from the expected value of 21.1 rad/(T· M) for SF-57 glass.

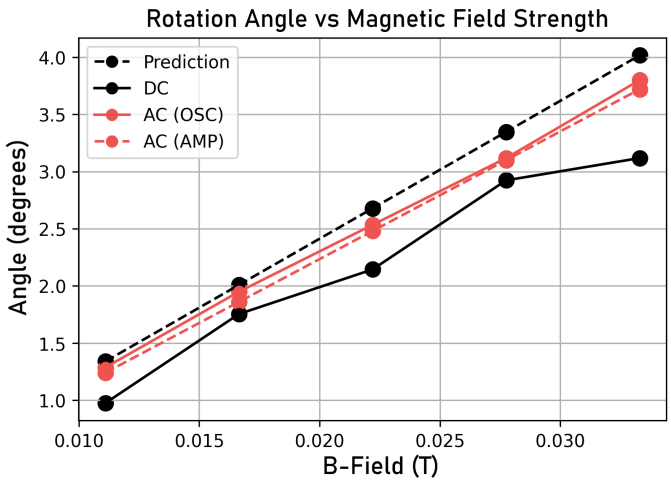

FIG. 4. Plot shows the angle of rotation of light after transmission through the SF-57 glass rod as a function of the applied magnetic field. The predicted rotation angle as determined from the expected Verdet constant of 21.1 is denoted by the dashed black line. Measurements from the DC (solid black line) and AC methods (red lines) are also shown.

E. The "Mystery" Material (Above and Beyond)

The experiment was repeated using an unknown "Mystery" glass rod (a glass rod labeled "Mystery") in place of the SF-57 glass rod. Since the Verdet constant of this material is not known beforehand, the same current configurations and measurement techniques were applied to determine its value, with the exception of the DC method. Driving the solenoid with direct current did not provide sufficient precision for accurately determining the angle of rotation.

The uncertainties resulting from previous results of each method were then used to evaluate the reliability of the new measurements. The most accurate estimate of the Verdet constant for the Mystery material was obtained and compared to known Verdet constants of various types of glass to identify the most likely candidate.

DC Method. Driving the solenoid with direct current resulted in inconsistent voltage readings from the photodetector, which fluctuated independent of the amount of applied current. An initial voltage of $V_0 = 0.130\ V$ was obtained in the absence of a magnetic field, but nevertheless, this method was deemed ineffective and not used for this material.

OSC Method. The photodetector signal obtained from applying an AC-field to the solenoid was plotted on the oscilloscope as a function of time. The peak-to-peak voltages for each current value, shown in Table III, were halved to obtain the signal amplitude and substituted into Eq. 2 along with V_0 to determine the Faraday rotation angle. Using Eq. 1, the Verdet constant for the mystery material obtained from the OSC method was found to have a mean value of 12.7 rad/(T·M).

Current (A)	Amplitude (V)	θ_{rad}	θ_{deg}	Verdet Const.
1.0	0.0038	0.014	0.83	13.0
1.5	0.0059	0.023	1.31	13.7
2.0	0.0070	0.027	1.54	12.1
2.5	0.0090	0.035	1.98	12.5
3.0	0.0106	0.041	2.34	12.2

TABLE III. "Mystery" Glass: Measured amplitude of oscillation signal and corresponding angle of polarization for different current values using an AC magnetic field. Verdet constants have units of rad/(T·M).

AMP Method. Alternating current was once again used to drive the solenoid and generate an oscillating magnetic field. The DC component of the signal was isolated using the lock-in amplifier while reducing noise to obtain output voltages for each value of current.

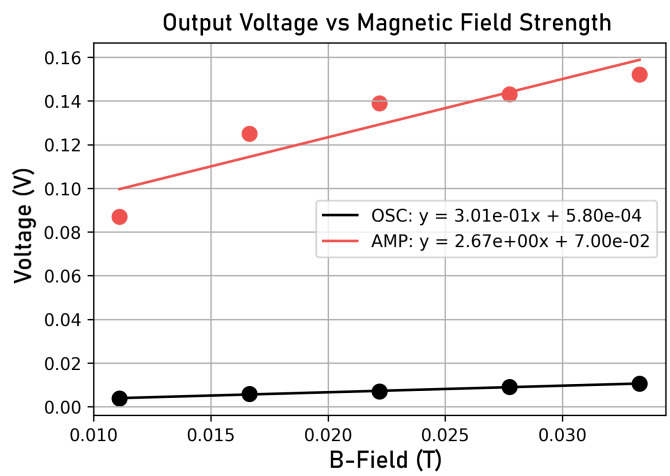

FIG. 5. "Mystery" Glass: Plot shows output voltage vs magnetic field strength for the two AC methods: Oscilloscope (OSC) and Lock-in Amplifier (AMP). Lines indicate the slopes obtained using regression analysis.

Using the slope values of 0.301 and 2.67 (Fig. 5) for the OSC and AMP method respectively, the scaling factor was found to be 0.113. After substituting this factor and the slope value of 2.67 into Eq. 5, the Verdet constant of the mystery material found using the AMP method was 11.6 rad/(T· m).

The Verdet constants of 12.7 and 11.6 obtained from the OSC and AMP method for the "Mystery" material were found to align closest with SF-5, a diamagnetic glass with a Verdet constant of 13.1 when using 633 nm light [3]. For the 650 nm laser diode, its Verdet constant as a function of wavelength was found to be 12.5. The constants of 12.7 and 11.6 deviate from this value by 1.6% and 7.2%, respectively, making SF-5 the most likely candidate. Mystery solved.

CONCLUSION

For the SF-57 glass rod, a Verdet constant of $\nu = 20.0$ rad/(T·m) was obtained from the oscilloscope (OSC), with an error of around 5% from the theoretical value of 21.1. The value of 19.5 obtained from the lock-in amplifier (AMP) was similar in accuracy, with an error of around 7.6%. Applying a DC magnetic field (DC) yielded a mean Verdet constant of 17.1, with an error of around 19% below the expected value.

For the SF-5 glass rod, a Verdet constant of $\nu = 12.7$ rad/(T·m) was obtained from the OSC method, with an error of around 1.6% from the theoretical value of 12.5. The value of 11.6 obtained from the AMP method was similar in accuracy, with a larger error of around 7.2%.

The OSC method was found to be the most accurate in determining the Verdet constants of both SF-57 and SF-5. Fig. 4, which plots rotation angle versus magnetic field strength, shows a similar trend line for both AC methods. However, the AMP method exhibited a more consistent slope relative to neighboring data points, indicating higher precision despite its slightly lower accuracy.

Theoretically, the lock-in amplifier with its phase-sensitive detection is expected to be more sensitive in comparison to the oscilloscope. However, the superior accuracy from the OSC method was likely due to a strong signal-to-noise ratio resulting from the setup. The advantages of frequency matching and signal isolation from the lock-in amplifier would have been more pronounced provided the amount of noise had sufficiently exceeded the detectable signal.

The DC method was the least precise method in determining the Verdet constants for both SF-57 and SF-5. For SF-57, the value obtained using the DC method was considerably low compared to the values obtained from the AC methods, making it the least accurate of the three methods used. For SF-5, the DC method was deemed ineffective due to inconsistent voltage readings that fluctuated independently of the magnetic field, making it the least precise method for this material. Since no reliable Verdet constant was extracted for SF-5 using this method, its accuracy could not be evaluated for this material.

The constant signal produced by the DC method is an inherent limitation to both its accuracy and precision. Any noise or drift resulting from the experimental setup leads to variations in the measured signal over time. This effect was significantly reduced using AC methods, which isolate the signal at a specific frequency and suppress drift and background noise.

The effects of Faraday rotation observed in this experiment demonstrated the significantly improved reliability of AC over DC current configurations. The oscilloscope method yielded the highest accuracy for both SF-57 and SF-5, while the amplifier method exhibited greater precision. Ultimately, the effectiveness of the experimental methods depends not only on the inherent accuracy and precision of the instruments, but also on how well the methods align with the characteristics of both the signal and experimental setup.

AUTHOR CONTRIBUTIONS

S.T. wrote the introduction and the experimental setup, G.D. and S.T. wrote the results, S.T. wrote the conclusion and formatted the final report.

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